

HW2

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$$\lim_{x \rightarrow \infty} x \cdot \sin \pi/x$$

$$= \lim_{x \rightarrow \infty} \frac{\sin \pi/x}{1/x}$$

L'Hospital

$$\downarrow$$
$$= \lim_{x \rightarrow \infty} \frac{(\sin \pi/x)'}{(1/x)'}$$

$$= \lim_{x \rightarrow \infty} \frac{(\cos \pi/x) \cdot (\pi/x)'}{-1/x^2}$$

$$\left[\text{note: } \left(\frac{\pi}{x}\right)' = \pi \cdot \left(\frac{1}{x}\right)' \right. \\ \left. = \pi \cdot \left(-\frac{1}{x^2}\right) \right]$$

$$= \lim_{x \rightarrow \infty} \frac{(\cos \pi/x) \cdot \pi \cdot (-1/x^2)}{-1/x^2}$$

$$= \lim_{x \rightarrow \infty} (\cos \pi/x) \cdot \pi$$

$$\left[\text{note } \frac{\pi}{\infty} = 0 \right]$$

$$= (\cos 0) \cdot \pi$$

$$= \pi$$

29 $\lim_{x \rightarrow 1} \underbrace{(2-x)^{\tan[(\pi/2) \cdot x]}}_A$

$$\ln A = \ln (2-x)^{\tan[(\pi/2) \cdot x]}$$

$$= \tan[(\pi/2) \cdot x] \cdot \ln(2-x)$$

$$= \frac{\ln(2-x)}{\cot[(\pi/2) \cdot x]}$$

$$\Rightarrow \lim_{x \rightarrow 1} \ln A = \lim_{x \rightarrow 1} \frac{\ln(2-x)}{\cot[(\pi/2) \cdot x]}$$

L'Hôpital

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{2-x} \cdot (-1)}{-\frac{1}{\sin^2(\pi/2 \cdot x)} \cdot \frac{\pi}{2}}$$

note:

$$[\cot(\pi/2 \cdot x)]' = -\frac{1}{\sin^2(\pi/2 \cdot x)} \cdot \pi/2$$

$$\lim_{x \rightarrow 1} \ln A = \frac{-1}{-1 \cdot \pi/2} = \frac{2}{\pi}$$

$$\Rightarrow \lim_{x \rightarrow 1} A = e^{2/\pi}$$

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$$\lim_{x \rightarrow 0^+} \underbrace{\left[-\frac{1}{\ln x} \right]^x}_A$$

$$\ln A = \ln \left[-\frac{1}{\ln x} \right]^x$$

$$= x \cdot \ln \left[-\frac{1}{\ln x} \right]$$

note: $\ln \frac{a}{b} = \ln a - \ln b$

$$= x \cdot \ln \left[\frac{1}{-\ln x} \right]$$

$$= x \cdot \left[\ln 1 - \ln(-\ln x) \right]$$

$$= x \cdot \left[0 - \ln(-\ln x) \right]$$

$$= -x \cdot \ln(-\ln x)$$

$$= \frac{-\ln(-\ln x)}{1/x}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \ln A = \lim_{x \rightarrow 0^+} \frac{[-\ln(-\ln x)]'}{(1/x)'} =$$

$$= \lim_{x \rightarrow 0^+} \frac{-\frac{1}{-\ln x} \cdot (-\ln x)'}{-1/x^2}$$

$$\text{not: } [\ln g(x)]' = \frac{1}{g(x)} \cdot g'(x)$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\ln x} \cdot -\frac{1}{x}}{-1/x^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{\ln x} \cdot \frac{1}{x} \cdot x^2$$

$$\lim_{x \rightarrow 0^+} \ln A = \lim_{x \rightarrow 0^+} \frac{x}{\ln x} = \frac{0}{-\infty} = 0$$

$$\Rightarrow \lim_{x \rightarrow 0^+} A = e^0 = 1$$